A Multi-Metric Evaluation and Comparison of Large Language Models for German Easy Language Translation

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Abstract

This study investigates the potential of large language models (LLMs) for intralingual translation into Easy German. We evaluate three LLMs for intralingual translation on administrative and medical texts on a number of lexical, syntactic, and readability complexity dimensions. In addition, we also perform statistical validation. Results show clear differences among the models. While Qwen shows a stronger ability for lexical simplification, Llama achieves more consistent syntactic simplification. GPT-5.1 has only limited advantages that are statistically supported. While lexical and structural differences are significant, global readability scores do not differ much across models. Overall, the findings indicate that current LLMs can effectively reduce linguistic complexity and thus be used for Easy Language translation. However, performance varies depending on the analyzed linguistic level and model design, which emphasizes the need for evaluation under the application of multiple metrics.

1 Introduction

Guided by recent federal accessibility acts[1], German public authorities and digital service providers are required to translate official notices and further documents into German Easy Language aimed at individuals with intellectual or psychological disabilities. However, this growing demand cannot be met by human translators alone, opening opportunities for the application of Artificial Intelligence (AI) in intralingual translation. These opportunities are even more promising given that Easy Language is a rule-based variety following strict rules on text, sentence, and word level (Bredel and Maaß, 2016). As a consequence, it is often regarded as a controlled language (Hansen-Schirra et al., 2020), which in turn means that, seen from a theoretical perspective, it offers a high automation potential. Still, while there are recent studies on the use of AI in intralingual translation, many aspects remain unexplored, e.g., text type and genre variation. In the present study, we compare the performance of three LLMs (GPT-5.1, Llama-3.3-70b-instruct and Qwen3-235b-a22b) for intralingual translation of administrative and medical texts. For a comprehensive evaluation, we employ a range of different linguistic complexity measures.

The remainder of this paper is structured as follows. Section 2 summarizes existing related work. In Section 3, we outline our research design and present the metrics we used for our analyses. Our results are then presented in Section 4, while we discuss some main findings and ideas for future work in Section 5.

2 Related Work

2.1 Easy German

Easy German is characterized by a maximally reduced complexity on all linguistic levels and is mainly intended for people with communication impairments. It aims to promote communicative inclusion and broader participation and has been a subject of scientific research since 2014 (Maaß et al., 2014). Later on, a scientifically grounded Easy German rule-book (Maaß, 2015), as well as the first empirical studies (Gutermuth, 2020; Deilen, 2021; Schiffl, 2022) have appeared.

In recent years, Easy German has also undergone increasing professionalization and standardization. Besides scientific rule books and applied translation guidelines, institutional standardization efforts such as the DIN SPEC https://www.din.de/en/innovation-and-research/din-spec-en for Easy Language have contributed to establishing more consistent linguistic and communicative

[1] Federal Act on Equal Opportunities of Persons with Disabilities (Behindertengleichstellungsgesetz [BGG] 2016/2018) and German Accessibility Improvement Act (Barrierefreiheitsstärkungsgesetz [BFSG] 2025)

criteria for accessible communication. This development reflects the growing societal and legal relevance of Easy Language in public administration, healthcare, and digital communication.

In recent years, studies on the use of AI for translating into simplified language varieties, such as Easy Language, have emerged (see 2.3).

2.2 Models for Easy German Language translation

Although we focus on evaluation of LLMs for intralingual tasks (i.e., translating from one language variety into another), studies aimed at developing or fine-tuning dedicated systems for Easy Language translation are also of relevance here.

Säuberli et al. (2020) adapt neural machine translation models to the specific linguistic characteristics of Easy German. Their results show that the models successfully learned to implement some essential features of Easy Language, such as selecting simple vocabulary, shortening long sentences, or deleting less relevant phrases. However, they also discuss some limitations of their models, such as bias or inaccuracies. Spring et al. (2021) expand the dataset from Säuberli et al. (2020) and propose a sentence-based machine translation method aimed at simplifying standard German texts according to various CEFR (Common European Framework of References for Languages) levels. The authors are able to reduce undesired copying behaviour of the system. Still, they conclude that there is still a need to refine the model's ability to distinguish between desired and undesirable sentence copies.

Anschütz et al. (2023) fine-tune LLMs on a corpus of monolingual Easy German texts. Their study reveals that the resulting fine-tuned models align more closely with the linguistic characteristics and structural norms of Easy German than the original models. However, they also conclude that the automatically generated Easy Language texts are not directly suitable for the intended primary target groups, as the underlying models are, for example, prone to hallucinations. Yet, whether pre-trained models without additional training and fine-tuning can successfully translate texts from standard German into Easy Language remains an open question.

Deilen et al. (2024) evaluate several models adapted or fine-tuned for German intralingual translation of medical texts into Plain and Easy German. Their study includes SUMM AI, an MT-supported translation system specifically designed for Easy and Plain Language communication. Unlike generalized LLMs such as ChatGPT, which are not specifically optimized for intralingual translation tasks, the SUMM AI models are fine-tuned using domain-specific parallel data, including human-produced intralingual gold-standard translations and Easy Language rules. The results show that models specifically adapted for intralingual translation and those fine-tuned with in-domain parallel texts outperform more general systems across several linguistic parameters.

While most existing studies focus on German Easy Language, similar approaches have also begun to emerge for other languages. Kuoraitė and Gružauskas (2025) also investigate how Lt-Llama-2-7b-hf, pre-trained on Lithuanian language data, can be adapted to the task of simplifying Lithuanian texts into easy-to-read language. They use specialized datasets to fine-tune the model and test the model with various prompts.

2.3 Evaluation of automatic translation

Some recent studies have assessed the feasibility of using LLMs for intralingual translation tasks. Degenhardt (2025) investigates ChatGPT and Mistral for intralingual translation from standard French into Easy French in the domain of sexual health communication. The study reports that LLM-generated Easy Language translations can achieve substantial reductions in readability and syntactic complexity, although limitations regarding content accuracy and consistency remain. These findings demonstrate that LLM-based intralingual translation approaches may generalize across domains and languages.

Alva-Manchego et al. (2025) present the results of the shared task on automatic text simplification based on the Common European Framework of Reference for Languages (CEFR). The results show that although LLMs perform well on this task, controlled simplification often requires complex and multi-iterative processes. The authors also point out that the existing automatic metrics do not always capture all aspects needed, and therefore, there is a need for reevaluation. For this reason, we decide to test several metrics which include not only pure readability scores but also consider further aspects of comprehension.

Some studies also explore the potential of prompting strategies. Briva-Iglesias and Gil (2025) explore the usefulness of GPT-4.0 for simplifying informed consent forms into plain language using various prompt strategies. The authors evaluate

the outputs with several readability metrics (Flesch Reading Ease, Gunning Fog Index, and SMOG Index). They show that a simple prompting strategy works better. A similar result is achieved by Papandreou et al. (2025) who also show that simple prompting can be as effective as a more complex one. Another study on intralingual translation into Easy German (Deilen et al., 2023) also tests prompt strategies with ChatGPT-3.5, i.e. a holistic and a linguistic one. The results vary depending on the quality measure used (correctness, readability, and syntactic complexity). However, the holistic (i.e., general translation instructions prompt without explicit operationalization of linguistic rules) also achieves promising results. Therefore, we decide to use a holistic prompting approach in our study too.

Recent research has increasingly explored the use of LLMs for accessibility-oriented text simplification and Easy Language translation. Schomacker et al. (2025) investigate the translation of German public authority web pages into German Easy Language using fine-tuned Llama-based models and parameter-efficient fine-tuning approaches. Their results suggest that specialized and fine-tuned open-source models can achieve promising simplification quality and, in some cases, outperform larger proprietary systems on selected readability metrics. At the same time, ethical implications of LLM-based text simplification have also received growing attention. Freyer et al.(2024) argue that automated accessibility technologies should be understood primarily as assistive rather than fully autonomous systems, emphasizing challenges related to transparency, reliability, audience appropriateness, and the risk of misinformation or oversimplification in accessibility-oriented communication.

3 Research Design

This section defines the design and analytical strategies that were used to evaluate the outputs produced by the three LLMs. We describe the data collection and annotation in 3.1. Section 3.2 is dedicated to the evaluation framework, which includes lexical, syntactic, and readability metrics, and additionally, statistical procedures used to compare these metrics.

3.1 Data collection

We collect texts from two publicly available datasets: administrative texts from (Deilen et al., 2023) and medical texts from (Deilen et al., 2024) [2] We produce intralingual translations from standard German into Easy German using a holistic prompt described by Deilen et al. (2023) and test the following models: the latest GPT model (GPT-5.1) as well as Llama-3.3-70b-instruct and Qwen3-235b-a22b [3]. We use a zero-shot instruction-based prompting strategy, in which the model is instructed to translate the input text into Easy German without providing examples or additional reasoning steps. The holistic prompt consists of a single instruction applied directly to the full input text (see Appendix 1). As stated in Section 2.3 above, the decision for a holistic prompting is based on previous studies and reflects a direct, full-text translation setting. Such an approach is consistent with human translation practice, where translators typically consider the entire discourse context rather than decomposing the text into structured sub-tasks.

The dataset under analysis contains 40 texts from two different text types (20 administrative and 20 medical texts). The content of administrative texts covers various citizen-related topics, including how to notify a change of address, how to change your last name, etc. The 20 medical texts contain health communication topics and are derived from the website of the German health magazine *Apotheken Umschau*.

To perform the analyses, we automatically process the data. For tokenization, morphosyntactic annotation, and dependency parsers, we use spaCy's German pipeline (de_core_news_sm).

3.2 Data analysis

For each source text, we compute a consistent set of indicators focusing on lexical complexity, readability, and syntactic complexity.

Lexical Complexity To evaluate lexical complexity, we apply three measures:

(1) Type-Token Ratio (TTR) computed as the number of unique word types divided by total token count after spaCy tokenisation. Punctuation and non-alphabetic tokens were excluded. All tokens were lowercased; see equation (1). Lower TTR values indicate greater lexical repetition, which is

[2] Dataset and analysis are available at https://github.com/happy522/A-Multi-Metric-Evaluation-and-Comparison-of-LLMs

[3] Intralingual translations were generated using GWDG: https://chat-ai.academiccloud.de/

consistent with Easy Language principles, emphasizing controlled vocabulary and repetition.

However, TTR is sensitive to text length and may artificially decrease in longer texts. We therefore interpret TTR values in combination with other lexical indicators (Lexical Density and Word Rarity).

$$\text{TTR} = \frac{\text{Number of Unique Words (Types)}}{\text{Total Number of Words (Tokens)}} \quad (1)$$

(2) Lexical Density (LD) computed as the proportion of content words (tokens whose POS tag is one of NOUN, VERB, ADJ, ADV) to the total number of alphabetic tokens. LD is a measure for estimating information load and complexity, with higher values indicating more information-rich content; while lower LD values indicate a lower proportion of content words and thus reduced information density; see (2).

$$\text{Lexical Density} = \frac{\text{Number of Content Words}}{\text{Total Number of Words}} \quad (2)$$

(3) Word Rarity (WR) was computed as the proportion of alphabetic tokens that are neither stop words nor represented by a word vector in the spaCy vocabulary. Tokens meeting both criteria were classified as rare, as the absence of distributional representations typically correlates with low corpus frequency and limited lexical coverage. A higher WR therefore reflects greater lexical sophistication and the use of less common vocabulary, whereas lower values indicate a preference for frequent, well-established lexical items that are more likely to be familiar to the target audience and thus support accessibility.

Readability Redability measures were evaluated based on

(1) Lesbarkeitsindex (LIX, Björmnsson, 1968) (readability index) calculated by the sum of two numbers: the average sentence length (number of words divided by number of sentences) and the percentage of long words (words with more than six characters) (as defined in (5)). Higher scores indicate more complex texts.

$$\text{ASL (Average Sentence Length)} = \frac{\text{Number of Words}}{\text{Number of Sentences}} \quad (3)$$

$$\text{PLW (\% of Long Words)} = \frac{\text{Number of Long Words}}{\text{Number of Words}} \times 100 \quad (4)$$

$$\text{LIX} = ASL + PLW \quad (5)$$

(2) Wiener Sachtextformel (WSF-4, Bamberger and Vanecek, 1982) - We implemented a computational variant of the Wiener Sachtextformel based on automatic tokenization with spaCy and a custom syllable-counting function (Pyphen for German). Four measures were derived: average sentence length (ASL), average number of syllables per word (ASW), the percentage of polysyllabic words with three or more syllables (P3), and the percentage of long words (words with more than six characters (P6)). These values were inserted into the WSF formula shown in 6. Lower scores indicate easier language, whereas higher scores reflect increased lexical and syntactic complexity.

$$\begin{aligned}\text{WSF} = {} & 0.1935 \cdot \text{ASL} + 0.1672 \cdot \text{P3} \\ & + 0.1297 \cdot \text{ASW} - 0.0327 \cdot \text{P6} - 0.875\end{aligned} \quad (6)$$

(3) Amstad's Flesch Reading Ease, German adaptation (FRE, Amstad, 1978) - The German adaptation of the Flesch Reading Ease formula was computed using automatic sentence segmentation (spaCy) and a custom syllable-counting function based on Pyphen for German hyphenation. Two variables were derived: average sentence length (ASL) and average number of syllables per word (ASW). The readability score was calculated according to Equation 7. Higher values indicate easier-to-read texts, while lower values reflect greater linguistic complexity.

$$\text{FRE}_{DE} = 180 - \text{ASL} - (58.5 \times \text{ASW}) \quad (7)$$

(4) Gunning Fog Index (Fog, Gunning, 1952) - The Gunning Fog Index was computed based on average sentence length (ASL) and the percentage of complex words within the text (see Equation 8). Sentences were segmented using spaCy, and complex words were defined as lexical tokens containing three or more syllables. In accordance with standard practice, stop words and named entities were excluded from the complex word count. The Fog Index estimates the years of formal education required to understand a text on first reading, with higher values indicating greater linguistic difficulty.

$$\text{Fog} = 0.4 \times (\text{ASL} + \text{PCW}) \quad (8)$$

PCW denotes the percentage of complex words (three or more syllables).

Syntactic Complexity Syntactic complexity is evaluated using three dependency-based measures: (1) Mean Dependency Distance (MDD) provided by Jing and Liu (2015) and based on Liu (2008) and Gibson (2000), computed as the average linear distance between each dependent token and its syntactic head in the dependency tree, as in (9). Only non-root and non-punctuation dependencies were included in the calculation. Larger MDD values indicate more complex sentence structures, as they require readers to maintain longer dependencies in working memory.

$$\text{MDD} = \frac{\sum|\text{position}_{\text{head}} - \text{position}_{\text{dependent}}|}{\text{Number of dependencies}} \quad (9)$$

(2) Subordination Index (SI). To compute the SI, finite verbs were first identified, as each finite verb typically corresponds to a clause. The number of subordinate clauses was then determined using dependency structure: finite verbs whose dependency label was not 'root' were treated as embedded (non-main) clauses. The subordination index was calculated as the proportion of subordinate clauses relative to all finite verbs; see (10). Higher SI values indicate greater use of embedded structures.

$$\text{SI} = \frac{\text{Number of Subordinate Clauses}}{\text{Total Number of Clauses (Finite Verbs)}} \quad (10)$$

(3) Dependency Crossings (Cross). To calculate dependency crossings, each word (dependent) was first linked to its governing word (head), and a crossing was identified when two arcs intersect when visualized linearly over the sentence. Only non-punctuation dependencies were included in the calculation. Higher dependency crossings values indicate more syntactic embedding or complex sentence constructions, while lower or zero values reflect linear, simple syntax typical of easy-to-read texts.

Statistical Validation We perform evaluation in two steps: (1) Comparing the outputs for the different text types (administrative and medical) within all three models, and (2) Comparing the outputs of GPT-5.1, Llama and Qwen for each text type.

For each metric, we first calculate the mean scores of generated administrative and medical Easy Language translations. These mean scores are then used as the basis for statistical comparisons.

For the statistical validation of differences between administrative and medical texts, we use Welch's independent-samples t-test (two-sided, unequal variances). We choose this test because the administrative and medical datasets come from separate and independent sources, so we cannot assume equal population variances. For each metric, we report the mean difference (administrative minus medical), its standard error, and a two-sided 95% confidence interval calculated using the Welch-Satterthwaite approximation. We measure effect sizes using Cohen's d based on the weighted pooled standard deviation, which gives a standardised measure of size. Because multiple linguistic metrics are evaluated simultaneously, we control for inflated Type I error by applying the Holm procedure to adjust p-values. The correction is applied across all evaluated metrics within each analytical comparison (i.e., lexical, syntactic, and readability metrics jointly tested per contrast), thereby controlling the family-wise error rate. Statistical significance is determined at α = 0.05 based on these adjusted values. Metrics that did not survive correction are interpreted as showing no reliable difference. For metrics that show significance, we interpret the direction of the effect based on a set criterion 3.2 that indicates whether higher or lower values represent better performance.

To compare GPT-5.1, Llama, and Qwen on the same input texts (n = 20), we use a repeated-measures non-parametric framework. We apply the Friedman test to evaluate overall differences across the three related model outputs for each metric. This design includes paired observations and does not assume normal distribution. We estimate effect sizes with Kendall's W to measure the strength of agreement in model rankings. When the Friedman test shows significant results (α = 0.05), we perform pairwise Wilcoxon signed-rank tests with Bonferroni correction to manage multiple comparisons. A metric is deemed statistically significant if at least one corrected pairwise comparison has $p < 0.05$. Since the task prioritizes accessibility, we define the performance direction in advance: lower values indicate improvement for complexity-based metrics (e.g., TTR, LD, WR, LIX, WSF, Fog, MDD, SI, cross), while higher values indicate improvement for readability-oriented metrics (e.g., FRE). This approach guarantees a reliable multi-model comparison while reducing Type I error and considering within-text dependency.

4 Results

In this section, we present the results of our evaluation along two dimensions: (1) Comparing the outputs for the different text types (administrative and medical) within all 3 models, and (2) Comparing the outputs of GPT-5.1, Llama and Qwen for each text type.

For each metric, we report mean values, the applied statistical test, significance levels, and effect sizes. Section 4.1 details the administrative vs. medical texts, while section 4.2 examines outputs of GPT 5.1, Llama and Qwen.

4.1 Administrative vs. Medical Texts

An overview of the results for administrative vs. medical texts is given in tables 3, 4, 5 in the appendix. Here, we illustrate the distribution of the metric scores across the models and group the scores into the three categories. Besides that, we also show the scores of the original text.

Lexical Measures As seen from Figure 1, the simplified versions turn out to be more complex than the original texts, as this is the case for TTR in the medical Llama outputs, LD in the medical GPT outputs, or WR in both administrative and Qwen outputs.

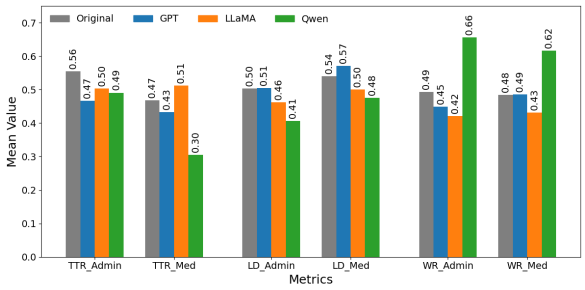


Figure 1: Lexical measure: Administrative vs. Medical text metrics

Across models, the clearest and most consistent lexical finding is Lexical Density (LD) administrative outputs are reliably less lexically dense than medical outputs in Qwen (admin = 0.406 vs. Med = 0.476; Δ = -0.069, adjusted p = .0063, d = -1.22), Llama (0.462 vs. 0.501; Δ = -0.038, adjusted p = .0041, d = -1.23), and GPT-5.1 (0.505 vs. 0.571; Δ = -0.066, adjusted p = .0020, d = -1.30). All three models show a large, practically important effect ($|d| \approx 1.2$–1.3, adjusted $p < .01$). Type–Token Ratio (TTR), by contrast, shows a very large domain difference only for Qwen (0.491 vs. 0.305; Δ = +0.186, adjusted $p < .001$, d = 2.43); Llama and GPT-5.1 show no reliable TTR difference after correction. Word Rarity (WR) shows small, non-replicable differences across models.

Readability Measures Figure 2 shows that the outputs of all systems are less complex than the original texts.

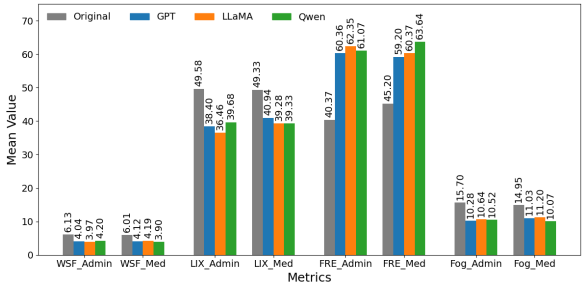


Figure 2: Readability measure: Administrative vs. Medical text metrics

Readability contrasts are weak and inconsistent across models, with no index surviving Holm correction 2. For Qwen, medical texts show slightly lower WSF (3.898 vs. 4.196; Δ = +0.298, d = 0.60) and Fog (10.520 vs. 10.072; Δ = +0.449, d = 0.39), and slightly lower FRE (61.070 vs. 63.642; Δ = -2.572, d = -0.53), but none remains significant after adjustment (all adjusted $p \geq .3865$), while LIX differences are negligible (39.679 vs. 39.330; Δ = +0.349, d = 0.11). For Llama, LIX shows a moderate unadjusted difference favoring easier administrative texts (36.456 vs. 39.280; Δ = -2.825, d = -0.67), yet this effect fails Holm correction (adjusted p = .3699), and all other metrics show only small effects ($|d| \leq 0.28$, adjusted p = 1.000). GPT-5.1 likewise shows only small, non-significant differences: LIX (38.399 vs. 40.939; Δ = -2.541, d = -0.39), Fog (Δ = -0.751, d = -0.32), FRE (Δ = +1.159, d = 0.10), and WSF (Δ = -0.074, d = -0.08), all with adjusted p = 1.000. Effect sizes are small to moderate, confidence intervals overlap zero, and no readability metric provides a robust or replicable distinction between administrative and medical outputs in this dataset.

Overall, Lexical Density is the only cross-model robust discriminator (large negative d in all three models); Qwen additionally shows pronounced model-specific lexical (TTR) and syntactic (crossings) contrasts. Readability metrics are not reliable indicators of domain differences here.

Syntactic Measures As seen from Figure 3, most of the outputs remain similarly complex to the original texts, at least as measured with MDD and SI.

The Cross score reveals a domain-dependent difference: while the medical outputs of all the models are considerably less complex, the administrative outputs of GPT and Qwen are even more complex than the underlying original texts.

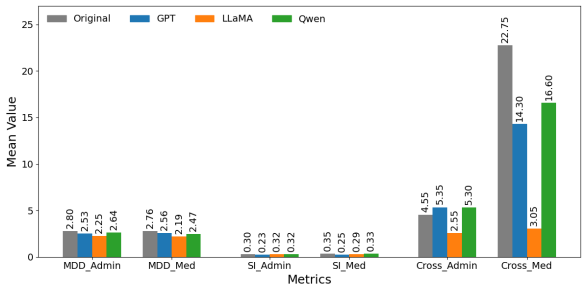


Figure 3: Syntactic measure: Administrative vs. Medical text metrics

Syntactic contrasts are largely model-specific 3. Dependency crossings are substantially lower in administrative than in medical outputs for Qwen (5.30 vs. 16.60; Δ = -11.30, adjusted p = .0007, d = -1.46), indicating much simpler syntactic configurations in Qwen's administrative texts; this effect does not survive correction for Llama (Δ = -0.50, adjusted p = 1.00) or GPT-5.1 (Δ = -8.95, adjusted p = .4079). Mean dependency distance (MDD) shows a moderate–large effect in Qwen (Δ = +0.170, d = 0.88) but fails Holm correction (adjusted p = .0585) and is null for Llama and GPT-5.1. Subordination Index (SI) shows no consistent differences across models.

In summary, only Qwen exhibits robust syntactic simplification (Crossings) by domain; other syntactic measures are inconclusive or only suggestive.

4.2 GPT-5.1 vs. Llama vs. Qwen

Table 1 presents the results of the comparison between GPT-5.1 vs. Llama vs. Qwen for administrative texts; Figure 4 represents mean value distribution across models.

Administrative Texts Table (1) indicates that across the administrative texts, most lexical measures show significant differences. Type-Token Ratio and Lexical Density both differ significantly (p = .0058). GPT-5.1 has the lowest TTR (M = 0.466), and Qwen has the lowest LD (M = 0.406). Word Rarity also shows a strong difference (p < .001), with Llama producing the lowest values (M = 0.420). This indicates that Llama uses more common vocabulary.

Looking at readability, only LIX shows a significant difference (p = .026). Llama has the lowest score (M = 36.456). No significant differences are found for WSF (p = .580), FRE (p = .387), and Gunning Fog (p = .848).

In syntax measures, Llama performs the best. It achieves the lowest Mean Dependency Distance (M = 2.255, p < .001) and has the fewest Dependency Crossings (M = 2.550, p < .001). This reflects simpler sentence structures. GPT-5.1 has the lowest subordination index (M = 0.231, p = .035), which indicates it uses fewer subordinate clauses.

Overall, Llama produces the simplest sentence structures, Qwen reduces lexical density, and GPT-5.1 primarily lowers subordination. 4

Medical Texts For the medical texts 2, we see clear differences in all lexical measures. The Type-Token Ratio (p < .001) and Lexical Density (p < .001) are lowest for Qwen (TTR = 0.317; LD = 0.483). This shows a more repetitive vocabulary with less content. Word rarity also differs significantly (p < .001). Llama has the lowest value (M = 0.449), which means it uses more common words.

For readability, LIX (p = .522) and FRE (p = .287) show no significant differences among the models. However, WSF (p = .047) and the Gunning Fog Index (p = .010) show significant differences. Qwen has the lowest value for both (WSF = 4.007; Fog = 10.358), indicating that the texts are easier to read.

On the syntactical level, Llama performs the best overall. It has the lowest Mean Dependency Distance (M = 2.205, p = .001) and the fewest Dependency Crossings (M = 2.350, p < .001), indicating simpler sentence structures. GPT-5.1 has the lowest Subordination Index (M = 0.246, p = .002), meaning it uses fewer subordinate clauses.

Overall, Llama stands out for simple syntax, Qwen for lower lexical complexity and readability, and GPT-5.1 for fewer subordinate clauses.

4.3 Errors

In this study, we have primarily focused on automatic evaluation metrics, and therefore, no detailed error analysis has yet been performed. However, a preliminary review of selected texts has revealed that the translations contain errors from different error categories that we illustrate below.

The sentence in (1-a) is grammatically incorrect because the conjugated form of the verb *to be* is missing in this sentence. (1-b) illustrates a missing

Metric	GPT-5.1_mean	Llama_mean	Qwen_mean	Friedman_p	Kendall's_W	Winner	Posthoc
TTR	0.466	0.504	0.491	0.005799	0.3681	GPT-5.1	Significant
LD	0.505	0.462	0.406	0.005799	0.4152	Qwen3-235B	Significant
WR	0.449	0.420	0.656	0.000032	0.3668	LLaMA-3.3-70B	Significant
WSF	4.045	3.971	4.196	0.580246	0.4952	LLaMA-3.3-70B	Not Significant
LIX	38.399	36.456	39.679	0.025991	0.5049	LLaMA-3.3-70B	Significant
FRE	60.362	62.345	61.070	0.386741	0.5569	LLaMA-3.3-70B	Not Significant
Fog	10.280	10.641	10.521	0.848269	0.5233	GPT-5.1	Not Significant
MDD	2.526	2.255	2.639	0.000184	0.2511	LLaMA-3.3-70B	Significant
SI	0.231	0.319	0.319	0.035084	0.3099	GPT-5.1	Significant
Cross	5.350	2.550	5.300	0.045175	0.3332	LLaMA-3.3-70B	Significant

Table 1: Comparison of GPT-5.1, LLaMA-3.3-70B, and Qwen3-235B across Linguistic Metrics (Administrative texts)

Metric	GPT-5.1_mean	Llama_mean	Qwen_mean	Friedman_p	Kendall's_W	Winner	Posthoc
TTR	0.433	0.503	0.317	0.000002	0.2945	Qwen3-235B	Significant
LD	0.571	0.511	0.483	0.000001	0.3338	Qwen3-235B	Significant
WR	0.485	0.449	0.617	0.000000	0.4957	LLaMA-3.3-70B	Significant
WSF	4.119	4.440	4.007	0.047359	0.5445	Qwen3-235B	Significant
LIX	40.939	41.131	40.457	0.522046	0.4784	Qwen3-235B	Not Significant
FRE	59.203	59.001	62.642	0.286505	0.5569	Qwen3-235B	Not Significant
Fog	11.031	12.015	10.358	0.009562	0.5195	Qwen3-235B	Significant
MDD	2.562	2.205	2.436	0.001294	0.4379	LLaMA-3.3-70B	Significant
SI	0.246	0.295	0.328	0.002358	0.4828	GPT-5.1	Significant
Cross	14.300	2.350	16.850	0.000003	0.5731	LLaMA-3.3-70B	Significant

Table 2: Comparison of GPT-5.1, LLaMA-3.3-70B, and Qwen3-235B across Linguistic Metrics (Medical texts)

article before the noun *integration course*. It also has some stigmatising features, which we discuss in (2).

(1) Grammar:
   a. *Du dort als Nebenwohnung gemeldet* [*You registered there as secondary residence*].
   b. *Sie können Integrationskurs machen, wenn Platz ist* [*You can take integration course if space is available*].

In (2-a), the phrase *wenn Platz ist* is too colloquial in this context. Also, the phrase *Sie können Integrationskurs machen* can be perceived as stigmatizing: it mirrors a grammatical error commonly made by German learners with low language proficiency (i.e., article omission). Therefore, the intentional use of wrong German is problematic. In (2-b), the reader is addressed informally, although the formal form would be more appropriate[4].

(2) Audience appropriateness:
   a. *Sie können Integrationskurs machen, wenn Platz ist* [*You can take the integration course if space is available*].
   b. *Du hast eine Zweitwohnung in Braunschweig* [*You have a secondary residence in Braunschweig*].

In addition, we find several violations against Easy Language Rules; see (3). (3-a) disregards the established ban on subordinate clauses in Easy Language. (3-b) shows compounds with three or more syllables that should have been segmented in easy language.

(3) Rule violations
   a. *Eltern, die im Landkreis Hildesheim wohnen, aber nicht in der Stadt, stellen den Antrag bitte im Landratsamt* [*Parents who live in the district of Hildesheim but not in the city should please submit the application at the district office*].
   b. *Krankenversicherungsnachweis* [*proof of health insurance*], *Asylbewerberleistungsgesetz* [*asylum seekers' benefits act*].

[4]In German, there is a distinction between informal and formal second-person pronouns: while *du* is used for informal address (with children, friends, or family), *Sie* is the formal form (professional or public contexts).

5 Discussion and Future Work

This paper evaluates three large language models, GPT-5.1, Llama-3.3-70B, and Qwen3-235B, for intralingual translation into Easy German. The LLMs are tested for two different text types (administrative and medical texts) using a simple, holistic prompting strategy. The results show differences in quality based on the model and the specific metrics. This indicates that the quality of simplification varies depending on the linguistic level and type of text, which is also consistent with previous research on automatic text simplification and Easy Language translation (Deilen et al., 2023), (Alva-Manchego et al., 2025). In some cases, automatic simplification fails or is not reflected in the results, as the models' outputs have more complex structures than the underlying source texts. Similar limitations regarding consistency and reliability have also been reported by (Degenhardt, 2025) and (Deilen et al., 2024).

When comparing administrative vs. medical Easy Language translations, we find that administrative texts have a simpler syntactic structure and a lower lexical density than medical texts. The comparison of GPT-5.1, Llama and Qwen reveals that Qwen performs best in terms of lexical simplification, whereas Llama performs best in terms of syntactical simplification. In addition to that, Qwen also achieves the highest readability scores. Except for subordination clauses, GPT-5.1 performs worse than Qwen and Llama.These findings support previous studies showing that different models exhibit strengths on different linguistic dimensions and that readability metrics alone are insufficient for evaluating accessibility-oriented simplification (Alva-Manchego et al., 2025)

Our analysis of selected texts reveals that the translations contain not only content-related mistakes but also several violations against established Easy Language rules. Similar issues, including hallucinations and audience-inappropriate formulations, have also been observed in previous studies on Easy Language translation with LLMs (Anschütz et al., 2023) and (Freyer et al.). However, the comparison of the models in terms of the correctness of the content is still pending. In our future work, we plan to conduct a detailed error analysis and classify the errors according to an established error scheme. We also plan to rank the errors based on severity levels. This will allow us to complement the automatic metrics by a more comprehensive human evaluation and to better assess the suitability of LLMs for accessibility-oriented communication.

Limitations

One of our study's limitations is that we have not yet conducted complete error analyses of all texts. Therefore, we are still unable to conclude which model and approach performed best in terms of correctness of the content. Also, we have not assessed error severity up to this point, which, however, is crucial for the overall evaluation, as different types of errors may have different impacts on the translation quality. While incorrect terminology or omissions of negation markers may significantly change the intended meaning of a sentence, punctuation or spelling mistakes do not affect text comprehension and can usually be regarded as less severe. We are also aware of the reproducibility problem, as LLMs are regularly updated and are changing, which also implies that results by other studies using our datasets may differ from those reported by us. Another limitation is that we only analyzed texts from two domains aimed at lay people, which means that our results cannot be generalized to other text types, such as more specialized expert texts. In future studies, we therefore also aim to include specialized, more complex texts (such as expert-expert texts) from different domains, which will allow us to investigate whether the models' performance depends on domains and/or text complexity. In addition, we plan to expand the study to other LLMs.

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Appendix

1 Prompt

Holistic approach: "Übersetze folgenden Text in die Deutsche Leichte Sprache:" ("Translate the following text into German Easy Language") + text.

Temperature: 0.5

2 Comparison of Administrative and Medical texts

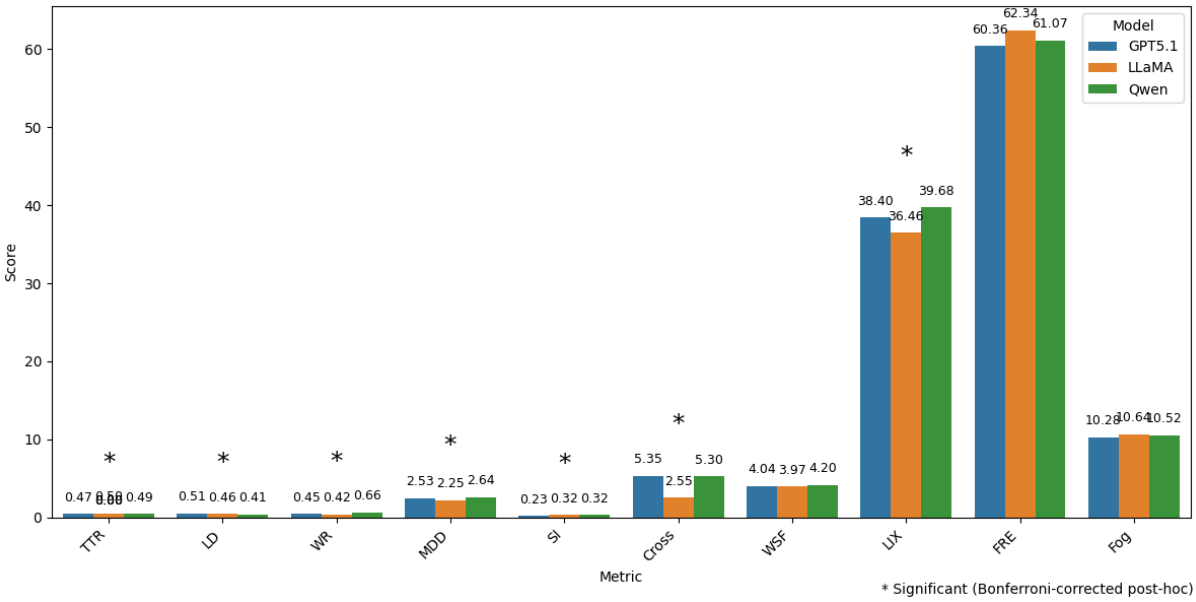


Figure 4: Comparison of GPT-5.1, Llama, and Qwen Across Linguistic Metrics (Administrative text)

Metric	Admin_mean	Medical_mean	Mean_difference	CI_lower	CI_upper	p_adjusted	Effect_size_d	Better
TTR	0.491	0.305	0.186	0.137	0.235	0.0000	2.433	Medical
LD	0.406	0.476	-0.069	-0.107	-0.032	0.0063	-1.220	Administrative
WR	0.656	0.616	0.039	-0.029	0.108	0.9209	0.378	No significance
WSF	4.196	3.898	0.298	-0.019	0.616	0.3865	0.602	No significance
LIX	39.679	39.330	0.349	-1.643	2.342	1.0000	0.112	No significance
FRE	61.070	63.642	-2.572	-5.690	0.546	0.5159	-0.528	No significance
Fog	10.520	10.072	0.449	-0.296	1.193	0.9209	0.386	No significance
MDD	2.639	2.469	0.170	0.046	0.294	0.0585	0.881	No significance
SI	0.319	0.330	-0.011	-0.051	0.029	1.0000	-0.172	No significance
Cross	5.300	16.600	-11.300	-16.311	-6.289	0.0007	-1.462	Administrative

Table 3: Comparison of Administrative and Medical texts Qwen

Metric	Administrative_mean	Medical_mean	Mean_difference	CI_lower	CI_upper	p_value	Effect_size_d	p_adjusted	Better
TTR	0.466	0.433	0.034	-0.042	0.110	0.3741	0.285	1.0000	No significant difference
LD	0.505	0.571	-0.066	-0.098	-0.033	0.0002	-1.302	0.0020	Administrative
WR	0.449	0.485	-0.037	-0.068	-0.005	0.0241	-0.743	0.2170	No significant difference
WSF	4.045	4.119	-0.074	-0.651	0.502	0.7951	-0.083	1.0000	No significant difference
LIX	38.399	40.939	-2.541	-6.713	1.632	0.2253	-0.390	1.0000	No significant difference
FRE	60.362	59.203	1.159	-5.988	8.306	0.7435	0.104	1.0000	No significant difference
Fog	10.280	11.031	-0.751	-2.271	0.769	0.3236	-0.316	1.0000	No significant difference
MDD	2.526	2.562	-0.036	-0.327	0.255	0.8033	-0.079	1.0000	No significant difference
SI	0.231	0.246	-0.015	-0.074	0.043	0.6052	-0.165	1.0000	No significant difference
Cross	5.350	14.300	-8.950	-17.942	0.042	0.0510	-0.653	0.4079	No significant difference

Table 4: Comparison of Administrative and Medical texts for GPT-5.1

Metric	Administrative_mean	Medical_mean	Mean_difference	CI_lower	CI_upper	p_value	Effect_size_d	p_adjusted	Better
TTR	0.504	0.512	-0.009	-0.048	0.031	0.6634	-0.139	1.0000	No significant difference
LD	0.462	0.501	-0.038	-0.058	-0.018	0.0004	-1.225	0.0041	Administrative
WR	0.420	0.432	-0.011	-0.036	0.014	0.3598	-0.293	1.0000	No significant difference
WSF	3.971	4.191	-0.220	-0.729	0.289	0.3866	-0.277	1.0000	No significant difference
LIX	36.456	39.280	-2.825	-5.529	-0.120	0.0411	-0.669	0.3699	No significant difference
FRE	62.345	60.370	1.975	-3.783	7.733	0.4910	0.220	1.0000	No significant difference
Fog	10.641	11.204	-0.563	-1.842	0.715	0.3776	-0.282	1.0000	No significant difference
MDD	2.255	2.185	0.070	-0.063	0.203	0.2961	0.335	1.0000	No significant difference
SI	0.319	0.292	0.028	-0.026	0.081	0.3033	0.330	1.0000	No significant difference
Cross	2.550	3.050	-0.500	-2.340	1.340	0.5854	-0.174	1.0000	No significant difference

Table 5: Comparison of Administrative and Medical texts for Llama